A Possible New Quantum Algorithm: Arithmetic with Large Integers via the Chinese Remainder Theorem

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Residue arithmetic is an elegant and convenient way of computing with integers that exceed the natural word size of a computer. The algorithms are highly parallel and hence naturally adapted to quantum computation. The process differs from most quantum algorithms currently under discussion in that the output would presumably be obtained by classical superposition of the output of many identical quantum systems, instead of by arranging for constructive interference in the wave function of a single quantum computer.

Quantum computation is still a solution in search of problems. To be profitably implemented on a (hypothetical) quantum computer, an algorithm presumably must satisfy two requirements:

1. It must be parallelizable, so that identical operations are to be performed simultaneously on different terms of a superposition state vector.
2. There must be a way of reading out the answer by an observation on the system — i.e., on the final state vector as a whole. Presumably this requires some kind of constructive interference of the terms in the superposition.

The second requirement is what distinguishes a quantum-computable problem from a generic parallelizable problem.

In the most intensively studied applications of quantum computing, factoring [1] and search [2], most of the quantum operations are devoted to building up an eigenstate (of the computer's basic observables) that represents the result of the computation. The required Hilbert-space maneuvering gives each application an air of specialness. The present paper proposes a quantum algorithm of a different kind, in which the constructive interference that builds up the answer takes place at the level of classical waves or signals. Arguably, eventual general-purpose quantum computation is more likely to be of this type.

Arithmetic by the Chinese remainder theorem is a highly parallel procedure [3–6]. The idea is that the fixed word size of a standard computer can be transcended by doing all addition, subtraction, and multiplication modulo each of a set of pairwise relatively prime integers (moduli), simultaneously. The string of integers (each less than its corresponding modulus) uniquely represents some nonnegative integer less than the product of all the moduli. Unlike standard digital arithmetic, the computations relative to each modulus are completely independent of each other (no carries or cross terms), so the process is immediately and completely parallelizable.

The algorithm for reconstructing the large integer in human-readable (decimal) notation is relatively complicated and should be avoided or postponed whenever possible. If the number is truly huge, even printing it out or transmitting it over a communications channel might be expensive, and the usefulness of its exact expression open to question. Instead, the advised strategy (in classical as well as quantum computation) is to do all computations with the large number internally, until a useful conclusion is reached. This conclusion may be qualitative, such as that the number is exactly zero. (For example, in quantum field theory the ratio of two such numbers may be the coefficient of some tensorial invariant in an effective Lagrangian. The theory may be renormalizable only if that particular term is actually absent.) Or, it may suffice to know the number in floating-point form, to a precision much less than the total number of digits.

We consider here some aspects of how Chinese-remainder arithmetic can be adapted to a quantum computer. The obvious strategy is to divide the qubits of the computer into three groups (data fields), one to hold the modulus, one to hold the result of a computation modulo that modulus, and one for work space (if needed). The initial state should be a sum over moduli of states wherein the first field holds the modulus and the other two are set to zero:

$$|\mathrm{i}\rangle = \sum_{j=1}^{R} |m_j\rangle|0\rangle|0\rangle.$$

The computation should then read the modulus and perform the calculation relative to that modulus, producing

$$|\mathrm{f}\,\rangle = \sum_{j=1}^{R} |m_j\rangle|u_j\rangle|0\rangle.$$

(The notation is that of [4] and [6].) We shall not discuss here how this initial state is to be prepared or how the calculation is to be performed; neither of these steps seems likely to be a major obstacle. (Quantum algorithms for the operations of modular arithmetic already appear in the literature — e.g., [7].)

The interesting and novel aspect of this quantum algorithm is the output procedure. We propose that the quantum apparatus can be made to emit a signal (optical or electronic) that is (for the jth pure state) a sequence of pulses with period m_j; a pulse occurs when (and only when) the time is (a basic time unit times) u_j mod m_j. The examined output signal will be the superposition of the signal from each term. Instead of arranging for constructive interference in the state of the qubits themselves and then measuring the qubits, therefore, one will look for constructive interference in the output signal. The strongest pulse will occur at time u, the number (less than $\prod m_j$) that is congruent to u_j mod m_j — that is, the answer!

The natural setting for this kind of procedure is a large ensemble of identical quantum computers. For example, in existing nuclear magnetic resonance experiments, each molecule is a computer. During "readout" each molecule independently collapses into some eigenstate, with corresponding signal emission. If more than one eigenstate is consistent with the computational algorithm, therefore, the output from the whole system is an expectation value of the basic observable, not an eigenvalue [8]. Thus in such a computation at least some of the superposition or interference takes place at the classical level, after the quantum observations have been performed. As explained in [8], although this feature is a nuisance in some computations (such as quantum search when more than one object satisfies the search criterion), it is quite desirable in others, where the sought information is a statistical property of a quantum state rather than the random outcome of a single measurement on it. The present situation is akin to the latter.

Before discussing practical difficulties of the proposed computation, let us look at an example (of course, with moduli too small and few to be of ultimate interest). Let $(m_1, m_2, m_3) = (3, 4, 7)$; this is essentially the circle-of-fifths example in [6], whose table the reader may find useful. The numbers represented range from 0 to 83. If $u = 42$, then $(u_1, u_2, u_3) = (0, 0, 2)$. The component of the signal from modulus 3 consists of pulses at $t = 0, 3, 6, \ldots$; that from modulus 4 of pulses at $t = 2, 6, 10, \ldots$; that from modulus 7 of pulses at $t = 0, 7, 14, \ldots$. The total signal (shown in Figure 1) has height 3 at $t = 42$, height 2 at 12 places, and height 1 at 36 places. The pattern has a nontrivial structure, which is symmetric about 42. (The pulses of height 2 fall into three classes, with respective periodicities 12, 21, and 28 — the pairwise products of the m_i .)

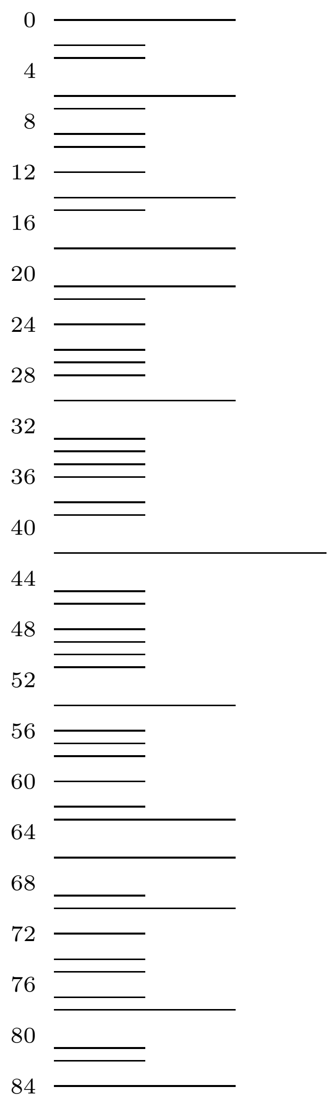

Fig. 1

There are two immediately obvious practical worries about this scheme. First, the bit pattern representing a number has length equal to the largest integer represented (*not* the number of digits in that integer, which is logarithmically smaller). By hypothesis, this number is very large. Second, if R, the number of moduli, is not very small, it may be hard to discriminate the pulse of height R representing the answer from the numerous pulses of height $R-1$. Fortunately, the pulse pattern for a given sequence of moduli is unique, and only its placement on the axis varies from one output number to another; furthermore, the pattern has such a structure that its center can still be located even if there is a slight uncertainty in some of the pulse heights. Therefore, to recognize the output number it is not necessary to recognize, or even to wait for, the highest pulse; one needs only to analyze enough of the output to recognize what part of the pattern it is coming from. (Implementation details obviously will require more input from number theory, pattern recognition, and experimental expertise.)

This observation, however, raises another concern: Since the pattern is always the same, the *phase* of the output must be measured with absolute precision in order to determine the answer exactly. But as previously remarked, quantum computation is no worse off than classical computation in this regard. The indicated experimental procedure should be able to recognize the answer to a decent number of digits of precision, relative to the size of the product of moduli. If the result of the computation is a qualitative conclusion that can be used internally by another kind of quantum computation, then the numerical readout procedure is not necessary at all.

It is difficult to be quantitative at this time about what capabilities for signal generation and detection would be necessary for a practical implementation of the algorithm: It is impossible to predict, for the time when serious quantum computers become reality, what will be the standard word size of a classical computer or the maximum size of an integer in a worthwhile computation.

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